



Fig. 3: Radiation field-regions of an antenna

Based on the reference levels set by ICNIRP, compliance distances from base-station antenna have been calculated with the help of equivalent isotropically-radiated power in the direction of maximum antenna gain (eirp, in watts) (Table II). The eirp is the product of the power supplied to the antenna and the maximum antenna gain relative to an isotropic antenna. It should always be assured that for distances greater than the compliance distance, the radiation level is under the limit. Thus, three types of exposure zones have been identified:

1. Compliance zone (potential exposure to EMF is below applicable limits)
2. Occupational zone (potential exposure to EMF is below the limits for occupational exposure, but exceeds the limits for general public exposure)
3. Exceedance zone (potential exposure to EMF exceeds the limits for both, occupational and general public exposure)

It should always be ensured that for distances greater than the compliance distance, the radiation level is under the limit. It means that if a lower amount of data concerning a radiating source is available, then the higher overestimation of compliance distances is required.

Assessment of exposure levels

The real source of electromagnetic radiation is the transmitting antenna and

not the transmitter itself. The transmitting antenna is the main source that determines electromagnetic field distribution in the vicinity of a transmitting station. So, it is required to consider the appropriate region of the field of an antenna.

The space surrounding an antenna is generally subdivided into four regions:

reactive near field, reactive-radiating near field, radiating (Fresnel) near field and radiating far field. Radial distance of the reactive near field is $0.62(D^2/\lambda)^{1/2}$, where D is the largest dimension of the antenna and λ is the wavelength (to be valid, D must be large compared to the wavelength). Radial limit of the radiating near field is $2D^2/\lambda$. The field region beyond radiating near field (distance $\geq 2D^2/\lambda$) is radiating far field, where the field pattern is essentially independent of the distance from the antenna. Thus, assessment of exposure level in any field region for mobile phone frequency bands of interest can be easily done.

For assessment of the exposure level, it is required to measure field quantities such as electric field intensity (E), magnetic field intensity (H) and specific absorption rate (SAR). Selection of the field quantity to be measured depends on where (near or far field region) the observer is and on the field impedance. In the reactive near field, it is required to measure both E and H components or evaluate SAR. Determining SAR instead of field measurement is preferable at positions located very close to the antenna. In reactive radiating near field, if no information on the field impedance is available, it is suggested to measure both E and H fields. If information on field impedance is available, it is possible to measure only one field component. If $E/H > 120\pi$ (high-impedance EMF), measurement of component

E is required and if $E/H < 120\pi$ (low-impedance EMF), measurement of component H will suffice.

Now, in radiating near field, it is suggested to measure only component E with the impedance taken equal to free-space impedance (Z_0). In the radiating far-field region, it is possible to measure either the electric or magnetic field component and determine the equivalent power density. However, measurement devices for the electric field component (E) are usually preferred. The equivalent power density within the far-field region is obtained from the measured field by calculation.

Further, multi-path reflections can create non-uniform field distributions. Therefore, to assess the correct exposure level, an averaging process is required and field values should be determined at several points (p) as shown in Fig. 4. Three points are basically recommended (Fig. 4a), but if accuracy is required the number can be increased to six (Fig. 4b), nine (Fig. 4c), 20 (Fig. 4d), etc, for more accurate results. In case of multiple sources, the measurement area should be divided into a grid of about one square metre and measurements should be performed at individual grid points.

The spatially-averaged field value is given by:

$$(E \text{ or } H) = \sqrt{\frac{\sum_{i=1}^n (E_i \text{ or } H_i)^2}{p}}$$

In most cases, a typical transmitting station contains many transmitting systems operating on many frequencies. In this scenario, all operating frequencies must be considered in a weighted sum, where each individual source is pre-rated according to the limit applicable to its frequency. For simultaneous exposure to fields at different frequencies of our interest (GSM, CDMA, 3G, WiMAX, etc), the compliance with exposure limits is evaluated as:

$$\sum_{i=1}^{10\text{MHz}} \left(\frac{E_i}{c} \right)^2 + \sum_{i=10\text{MHz}}^{300\text{GHz}} \left(\frac{E_i}{E_{L,i}} \right)^2 \leq 1$$

where, E_i is the electric field strength